Basis-space Spectral Path Tracing

Complete Macro-Optics Formulation

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SIGGRAPH '26 Supplementary Reference — Operator Algebra & Emitter Taxonomy

Spectral State & Hilbert Space Foundation

Every ray in BsSPT carries a full spectral state $\tilde{\alpha} \in \mathbb{R}^M$, $M = KN = 88$, representing radiance as a coefficient vector in the GHGSF basis. A spectrum $f(\lambda)$ is approximated as:

$$f(\lambda) \approx \sum_{i=1}^{M} \alpha_i \, b_i(\lambda) = \alpha^\top B_{:,\lambda}$$

The GHGSF basis uses $K = 11$ Gaussian-Hermite lobes, $N = 8$ modes per lobe, centers λ_k uniform on $[380, 830]$ nm:

$$b_{k,n}(\lambda) = \frac{H_n\left(\frac{\lambda-\lambda_k}{\sigma_k}\right) e^{-(\lambda-\lambda_k)^2/(2\sigma_k^2)}}{\eta_{k,n}}$$

The Gram matrix $G_{ij} = \langle b_i, b_j \rangle \approx (B \, \mathrm{diag}(w) \, B^\top)_{ij}$ has condition $\kappa(G) \approx 10^4$.

Whitening Transform

The Cholesky factorization $G = LL^\top$ defines the whitening map:

$$\tilde{\alpha} = L^\top \alpha, \qquad \tilde{B} = L^{-\top} B \in \mathbb{R}^{M \times L}$$

Whitened Gram: $\tilde{B} \, \mathrm{diag}(w) \, \tilde{B}^\top = I_M$. Every Galerkin operator with transfer $T(\lambda)$ becomes:

$$\boxed{\hat{A} = \tilde{B} \, \mathrm{diag}(w \odot T) \, \tilde{B}^\top}$$

G^{-1} vanishes from all render-time operations. Condition number drops $10^4 \to 1$.

Transport Architecture

The macro-optics formulation partitions the rendering problem into three categorically distinct roles:

Role	Acts on	Timing
Emitter	produces $\tilde{\alpha}_0$	load-time
Operator	transforms $\tilde{\alpha}$	render-time
Sensor	extracts scalar	render-time

A path of L bounces evolves as:

$$\tilde{\alpha}_L = \hat{A}_{e_L} \cdots \hat{A}_{e_2} \, \hat{A}_{e_1} \, \tilde{\alpha}_0$$

The full Monte Carlo estimator is:

$$\hat{I} = \frac{1}{N} \sum_{m=1}^{N} \beta_L^{(m)} \cdot \tilde{y}^\top \tilde{\alpha}_L^{(m)}$$

where β_L is the scalar geometric throughput and $\tilde{y} \in \mathbb{R}^M$ is the sensor function (e.g. CIE Y) projected into whitened space. All randomness is confined to geometric sampling; spectral evolution is *deterministic* Galerkin transport.

Operator Fusion

Operators acting on the same bounce compose via matrix multiplication:

$$\hat{A}_{\text{fused}} = \tilde{B} \, \mathrm{diag}(w \odot T_1 \odot T_2) \, \tilde{B}^\top$$

Example: Beer-Lambert absorption fused with Fresnel reflectance on a single bounce costs identical to either alone.

Emitter Taxonomy

Emitters are *not* operators. They produce the initial spectral state by Galerkin projection of a source SPD, executed once at scene load:

$$\tilde{\alpha}_0 = L^\top \cdot \mathrm{project}(\mathrm{SPD}(\lambda))$$

At render time, $\tilde{\alpha}_0$ is a precomputed constant vector — zero additional cost.

Planck's Emitter (Blackbody)

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{e^{hc/(\lambda k_B T)} - 1}$$

Named after Max Planck. The canonical thermal emitter; parameterized by temperature T in Kelvin. One projection per temperature value at load time. Covers incandescent bulbs, fire, stellar emission, any thermal source.

Generic SPD Emitters

All other sources reduce to the same projection pipeline with different SPD(λ):

- *Measured emitter* — tabulated spectroradiometric data (LEDs, fluorescent tubes, displays)
- *Sky emitter* — Hosek-Wilkie or Nishita model evaluated per sky state; solar disk treated as Planck's Emitter at $T \approx 5778$ K
- *Laser / monochromatic* — near-δ SPD; projects to a sparse $\tilde{\alpha}_0$ concentrated in lobes near the emission wavelength

Planck's Emitter earns physicist attribution because Planck's law is the SPD — attribution is exact. Other emitters are named by their model, consistent with the operator naming convention.

The Eight Spectral Operators

All operators are expressed in whitened space as $\hat{A} = \tilde{B}\,\mathrm{diag}(w \odot T)\,\tilde{B}^\top$ or a structure derivable from it. Energy stability requires $|\lambda_i(\hat{A})| \leq 1$ for all passive materials.

1. Beer-Lambert's Operator (Absorption)

Physics: $T(\lambda) = e^{-\sigma_a(\lambda)\,d}$, volumetric attenuation.
Structure: Near-diagonal, full rank, $O(M)$ apply.
Galerkin: $\hat{A} = \tilde{B}\,\mathrm{diag}(w \odot e^{-\sigma_a d})\,\tilde{B}^\top$
Materials: Colored glass, water, fog, participating media, atmospheric windows (notch variant of $T(\lambda)$).
Note: Atmospheric absorption bands (ozone, CO_2, H_2O) are a parameterization of this operator with piecewise-notched $T(\lambda)$ — no separate operator class required.

2. Fresnel's Operator (Surface Reflectance)

Physics: Schlick approximation $F(\theta, \lambda) = F_\infty(\lambda) + (1 - F_\infty(\lambda))(1 - \cos\theta)^5$.
Structure: Dense, full rank, $O(M^2)$; Schlick decomposes into four precomputed sub-operators:

$$\hat{A}_F(\theta) = P_0 + (1 - \cos\theta)^5 (I_M - P_0)$$

$P_0 = \tilde{B}\,\mathrm{diag}(w \odot F_\infty)\,\tilde{B}^\top$	normal-incidence
$P_{\mathrm{sq}} = \tilde{B}\,\mathrm{diag}(w \odot F_\infty^2)\,\tilde{B}^\top$	squared profile
$R_{\mathrm{cross}} = \tilde{B}\,\mathrm{diag}(w \odot F_\infty(1 - F_\infty))\,\tilde{B}^\top$	cross term
$Q_{\mathrm{comp}} = \tilde{B}\,\mathrm{diag}(w \odot (1 - F_\infty)^2)\,\tilde{B}^\top$	complement sq.

Materials: All dielectrics, metals, pigments; drives state-dependent MIS via scalar quadratic forms A, B, C computed in whitened space — no λ round-trips at render time.
Note: Kubelka-Munk pigment reflectance is a material model producing $R(\lambda)$; it feeds $F_\infty(\lambda)$ — same operator, different parameterization.

3. Fabry-Airy's Operator (Thin-Film Interference)

Physics: Path difference gives wavelength-selective reinforcement:

$$T(\lambda) = \tfrac{1}{2}\big(1 + \cos(4\pi n d/\lambda)\big)$$

Structure: Genuinely dense — all M singular values significant (rank > 80 for typical films). $O(M^2)$ unavoidable.
Materials: Soap bubbles, oil films, iridescent beetle wings, anti-reflection coatings, multilayer paints.

4. Stokes' Operator (Fluorescence)

Physics: Absorb at λ_a, emit at longer λ_e (Stokes shift $\Delta\lambda \approx 70\,\mathrm{nm}$). Bispectral kernel $K(\lambda_o, \lambda_i) = e(\lambda_o) \cdot a(\lambda_i)$ is rank-1 by separability.
Galerkin:

$$M_{\mathrm{raw}} = (Bw_e)(Bw_a)^\top \xrightarrow{\text{whiten}} \hat{A} = \tilde{e}\,\tilde{a}^\top$$

Apply: $\tilde{\alpha}' = (\tilde{a}^\top \tilde{\alpha})\,\tilde{e}$ — one dot product, one scale. $O(M)$.
Materials: Fluorescent dyes, optical brighteners, bioluminescent materials, certain fabrics.
Note: Raman scattering has a shifted-band kernel (nonzero near $\lambda_o = \lambda_i + \Delta$) — a banded off-diagonal generalization of this structure, constituting a distinct operator class for spectroscopy materials (*Raman's Operator*).

5. Cauchy's Operator (Dispersion)

Physics: $n(\lambda) = A + B/\lambda^2 + C/\lambda^4$ — refractive index varies with wavelength, routing different wavelengths to different geometric directions.
Structure: K rank-1 lobes (one per GHGSF lobe), softmax-normalized to enforce partition of unity:

$$\sum_{k=1}^{K} \hat{A}_k = I_M \quad \text{(partition of unity)}$$

Apply: $O(KM)$, $K = 11$. Spectral-geometric coupling: lobe k selected with probability s_k, operator $\hat{A}_k$ applied.
Materials: Prisms, optical glass, gemstones (diamonds), lenses, any dispersive dielectric.

6. Rayleigh's Operator (Molecular Scattering)

Physics: $\sigma_s(\lambda) \propto \lambda^{-4}$, dominant at short wavelengths (blue sky).
Structure: Identical skeleton to Beer-Lambert:

$$T(\lambda) = e^{-\sigma_s^{(R)}(\lambda)\,d}, \quad \sigma_s^{(R)} \propto \lambda^{-4}$$

Near-diagonal, gently banded toward short λ due to strong wavelength selectivity. $O(M)$.
Materials: Atmosphere, clear sky, any Rayleigh-scattering medium.

7. Mie's Operator (Particle Scattering)

Physics: Weak wavelength dependence for particles comparable in size to λ: $\sigma_s(\lambda) \propto \lambda^{-\alpha}$, $\alpha \in [0, 2]$.
Structure: Beer-Lambert skeleton with shallower $T(\lambda)$ profile. Near-diagonal with slight smoothing. $O(M)$.
Materials: Clouds, fog, smoke, haze, milk.

8. Raman's Operator (Inelastic Scattering)

Physics: Small wavelength shift due to vibrational modes: $K(\lambda_o, \lambda_i)$ concentrated near $\lambda_o = \lambda_i + \Delta$.
Structure: Banded off-diagonal matrix; structurally distinct from all other operators. Low effective rank (bandwidth $\ll M$). $O(M)$.
Materials: Spectroscopy-grade materials; semi-precious stones; certain biological tissues.

Completeness & Structural Properties

Categorical Completeness

Every physical light-matter interaction at the wavelength scale has exactly one home in this formulation:

Phenomenon	Classification
Thermal emission	Planck's Emitter
Measured SPD	Generic Emitter
Volume absorption	Beer-Lambert's Op.
Surface reflectance	Fresnel's Op.
Thin-film iridescence	Fabry-Airy's Op.
Fluorescence	Stokes' Op.
Chromatic dispersion	Cauchy's Op.
Molecular scattering	Rayleigh's Op.
Particle scattering	Mie's Op.
Inelastic shift	Raman's Op.
Atm. absorption bands	Beer-Lambert (notch T)
Pigment reflectance	Fresnel (K-M $R(\lambda)$)
Subsurface scattering	Outside scope†
Spectral caustics	Geometric β + Cauchy

†BSSRDF couples spatial positions — incompatible with per-ray $\tilde{\alpha}$.

Key Structural Properties

- **Zero spectral variance.** All randomness confined to geometric $\beta_L^{(m)}$. Spectral evolution is deterministic.
- **Operator fusibility.** Any set of operators on the same bounce fuses into a single $\hat{A} = \tilde{B}\ \mathrm{diag}(w \odot \prod_k T_k)\ \tilde{B}^\top$. One matmul regardless of how many effects are active.
- **Energy stability.** $|\lambda_i(\hat{A})| \leq 1$ for all passive materials — guaranteed by construction for diagonal-family operators; verified empirically for Fresnel and Fabry-Airy.
- **Symmetry.** All five canonical operators are symmetric ($\hat{A}^\top = \hat{A}$) — free backward pass for SsMLT sensitivity vectors.
- **Partition of unity.** Cauchy's Operator satisfies $\sum_k \hat{A}_k = I_M$ exactly in whitened space (softmax normalization).
- **Rank exactness.** Stokes' Operator is exact rank-1, not an approximation — physics-grounded by separability of the bispectral kernel.

Render-Time Cost Summary

Operator	Apply Cost	Structure
Beer-Lambert	$O(M^2)$ matmul	near-diagonal
Fresnel	$O(M^2)$ matmul	dense
Fabry-Airy	$O(M^2)$ matmul	dense
Stokes	$O(M)$ dot+scale	rank-1
Cauchy	$O(KM^2)$	K rank-1 lobes
Rayleigh	$O(M^2)$ matmul	near-diagonal
Mie	$O(M^2)$ matmul	near-diagonal
Raman	$O(M^2)$ matmul	banded

$M = 88$, $K = 11$. All matmuls are standard $M \times M$ gemv on precomputed $\hat{A}$.

SsMLT Extension

The Spectral-State Metropolis Light Transport algorithm (SsMLT) exploits the deterministic spectral evolution for a free gradient oracle. Given forward-pass operators $\{\hat{A}_k\}$, the sensitivity vector at vertex k is:

$$s_k = \left(\hat{A}_L \cdots \hat{A}_{k+1}\right)^\top \tilde{y}$$

computed via backward pass (one $M \times M$ matvec per vertex). Since all canonical operators are symmetric, $\hat{A}^\top = \hat{A}$ — no additional storage or computation. Stokes' and Raman's transpose swaps the role of emission and absorption vectors.

The MH target scalar is:

$$f(x) = \beta_L \cdot \tilde{y}^\top \tilde{\alpha}_L$$

The same luminance projection $\tilde{y}$ used for Russian Roulette — zero additional render cost. The sensitivity vectors drive spectral-aware mutation proposals: perturb geometry where $\|s_k\|$ is large to maximize spectral contrast in the proposed path.

All operator definitions, Galerkin derivations, whitening proofs, and CUDA implementation specifications are detailed in the BsSPT Operator Algebra Complete Reference and the BsSPT Operator Implementation Roadmap. This document is the macro-optics summary layer only.